SNP CrystalBridge

Understanding Guide
ECC to S/4HANA Data Migration

What it is | Why it works this way | How the pieces fit | How to decide | Self-check

AUDIENCE	BASED ON	COMPANION DOCUMENTS
New and existing members of the Sempra migration team	Kyano MC Guide, User Guide 25.09, Module 5 v25.03, TC 19.03 decks	Reference PDF (detail) and Worked Examples PDF (practice)

Prepared: October 2026 | Sempra Migration Team | Version 1.0

How to read this guide. Product facts (tools, reports, defaults, behaviour) come from the SNP documents. Everyday analogies are marked **Analogy** - they are teaching aids, not SNP statements. Examples with numbers are marked **illustrative**. Anything about the Sempra landscape is **TBC** and has not been assumed. Read this guide first, then use the Worked Examples to practise, and the Reference PDF to look up detail.

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01 The Big Picture

What CrystalBridge does and why it is built this way

In one sentence: **CrystalBridge copies the selected business data out of the old SAP ECC system, changes it on the way so it fits the new S/4HANA system, loads it, and then proves that it arrived correctly.** The ECC source is read, never modified. All changes happen between the export and the import.

1.1 Analogy: moving house

Analogy (not from SNP). Think of a house move. You decide what goes, pack it, store the boxes safely, relabel and place things in the new house, and finally check nothing is missing. The migration has the same five stages.

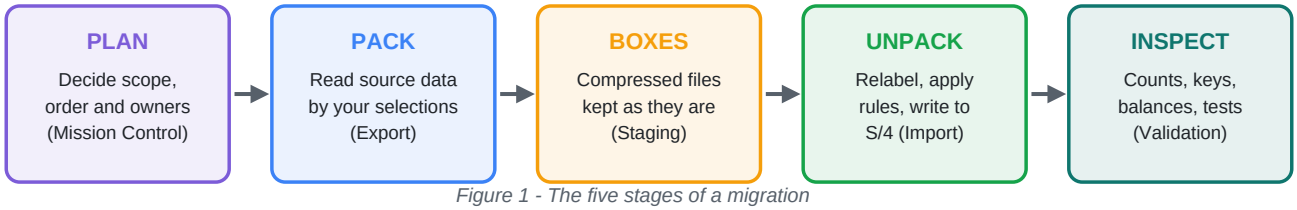

Figure 1 - The five stages of a migration

Moving house (analogy)	In CrystalBridge	Why it matters
Inventory of what to take	Selections and data objects	Defines the scope; wrong scope means wrong data.
Packing the boxes	Export block	Source stays untouched, so you can repeat the export safely.
Boxes in the garage	Export packages (compressed files)	Files are never overwritten; a repeat run adds new files.
New labels on the boxes	Mapping and rules	Company code 1000 becomes 2000; formats and keys are adapted.
Placing items in the new house	Import block	The import mode decides what happens if an item already exists.
Checking the inventory list	Validation (DCV, balances, tests)	Evidence for sign-off.

1.2 Five questions that explain the whole tool

Question	Answer in CrystalBridge	Where you work
What moves?	Whatever your selections and data objects describe, for example company codes and fiscal years.	Selections, Data Objects tabs
How does it change on the way?	Mapping tables and rules are applied between export and import.	Mapping tab, Rules
Where does it wait?	In compressed export files (staging), not directly between systems.	File system, Systems tab
How do I run it?	As runs made of blocks (export, import ...), by hand or from Mission Control.	Runs and Blocks tabs
How do I know it worked?	Logs, record counts, foreign key checks, balance comparison, application tests.	Log, DCV, validation tools

Why this design is safe. Because the source is only read, the export is repeatable. Because files are never overwritten, every run is traceable. Because rules sit between export and import, you can correct a rule and re-import without touching the source again.

02 Who Does What

The systems and the two main tools

2.1 Four places in the landscape

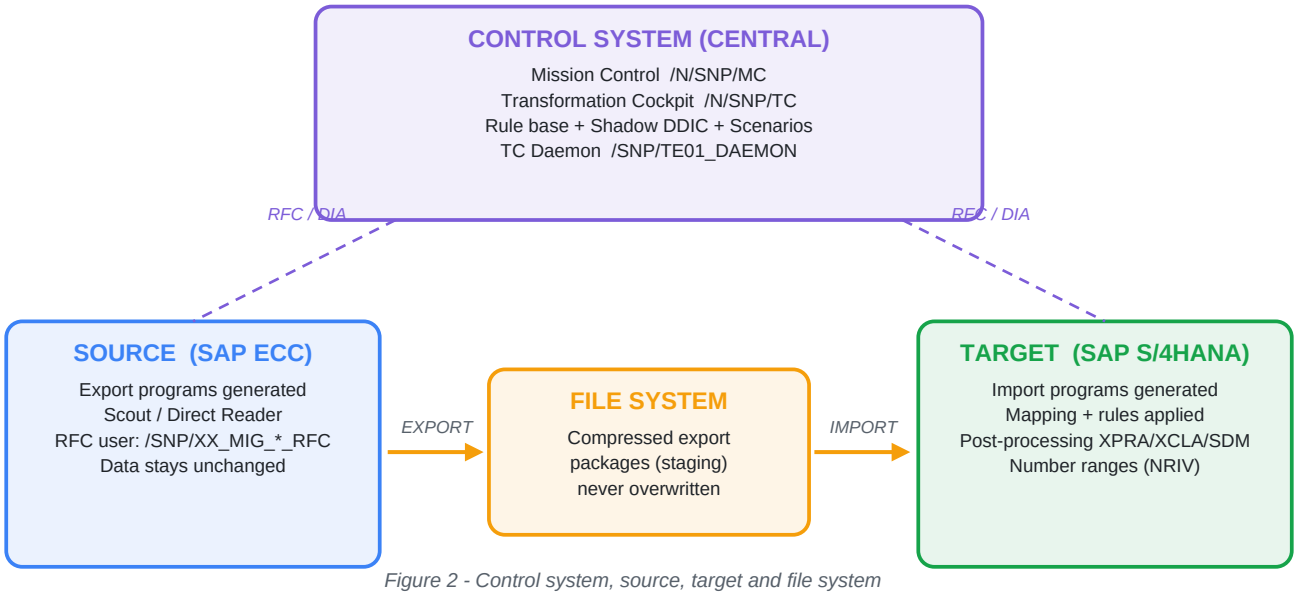

Figure 2 - Control system, source, target and file system

Place	Think of it as	What happens there	Typical contact
Control system	Command centre	Hosts Mission Control and the Transformation Cockpit, the rule base and the shadow DDIC. Generates the export and import programs and watches them run.	Migration lead, technical consultant
Source (SAP ECC)	The old house	Generated export programs read the data. Nothing is changed.	Basis, functional team
Target (S/4HANA)	The new house	Generated import programs write the data; mapping and rules are applied; post-processing runs here.	Basis, functional team
File system	The garage	Holds the compressed export packages between export and import.	Basis

The systems talk to each other through RFC connections. Every connection must be on the RFC whitelist of all systems and clients (maintained in `/SNP/TCSYS`). Sempra values for system IDs, clients and hosts are **TBC**.

2.2 Mission Control and the Transformation Cockpit

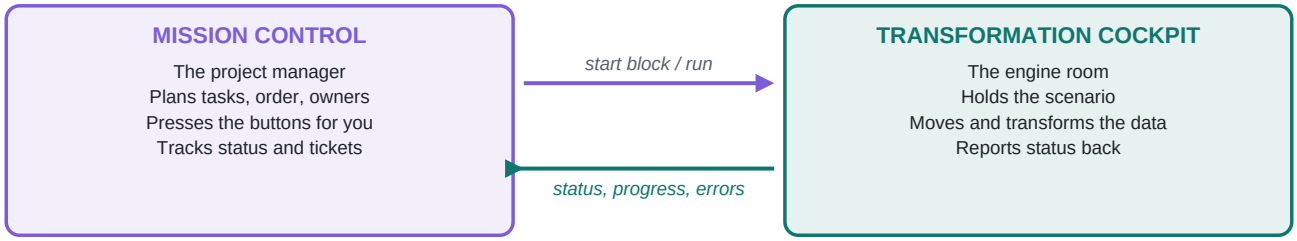

Figure 3 - The plan and the engine

	Mission Control (/N/SNP/MC)	Transformation Cockpit (/N/SNP/TC)
Main job	Plan and orchestrate the project: tasks, dependencies, owners, templates (units).	Do the data work: scenarios, selections, mappings, rules, runs and blocks.
Unit of work	A task (manual, report in dialog, report in background, or summary).	A block inside a run.
Status	Task status and tickets; a task is Blocked until its tickets are resolved.	Run and block status such as Running, Completed or Crashed.
Needs	A licensed project; no extra transports or RFC settings.	Transports and RFC whitelist on the systems.

How they connect. There is no magic link: a Mission Control task contains a small configuration that calls a cockpit API report (for example `/SNP/TE02_API_EXECUTE_BLOCKS`) and passes the scenario ID from the project variable `TC_SCENARIO`. A refresh class reads the cockpit status back into the task and can create tickets automatically.

2.3 The other CrystalBridge parts in one line each

Part	One-line meaning
CrystalBridge Analysis (AN)	Looks at the old system before the project: structure, usage, customizing, custom code, interfaces.
Data Consistency Verification (DCV)	After the load, checks foreign keys and domain fixed values to find broken references.
Rule-Based Near-Zero Downtime (RBNZD)	Moves most data while the source is live and only a small last delta during the cutover.
Shell Creator (SHC)	Creates the system shell used as basis for the target in the Bluefield approach.

03 The Building Blocks

Scenario, run, block, data object, table

Everything in the cockpit sits inside the next bigger thing. Learn this nesting once and the screens make sense.

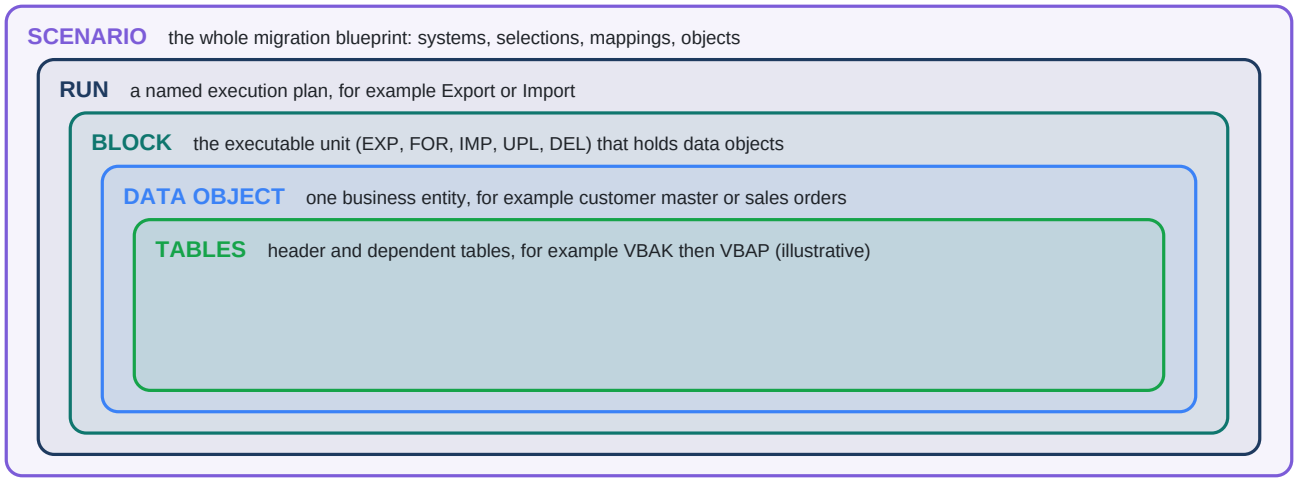

Figure 4 - The nesting from scenario down to tables

Level	What it is	What you decide here	Typical mistake
Scenario	The complete blueprint, identified by a unique ID of up to 30 characters.	Source and target systems, scope, which objects.	Reusing one scenario for unrelated purposes.
Run	A named execution plan.	Which blocks belong together and in which order.	Starting the wrong run in the wrong system.
Block	The unit that actually executes. Types: EXP export, FOR forerunner, IMP import, UPL upload, DEL deletion.	Parallel processes, package size, import mode, sequence.	Forgetting that import uses the last export unless you pick another.
Data object	A business entity at table level with its table hierarchy, for example sales orders.	Which objects are in the block; per table the export method and import mode.	Skipping the table analysis first.
Table	One DDIC table inside the object, header or dependent.	How it connects to its parent, selections, rules.	Wrong connection key, so wrong rows are read.

3.1 Header and dependent tables

A data object is a small family tree of tables. The **header (start) table** is selected by your scope; each **dependent table** is found through its key connection to the parent. This is why a correct selection on the header automatically brings the right items along.

Concept	Plain explanation	Detail
Start table	The table your selections apply to.	Dependent tables read relative to it.
Connection	The key fields that link child rows to parent rows.	Maintained per table in the data object.
Scout Reader (SR)	Default. Reads dependent rows for the keys already exported (FOR ALL ENTRIES).	Dependent tables start after the parent and use its last export.
Direct Reader (DR)	Reads with a join instead; independent of the parent export.	Starts with exactly one process; a restart means a full re-export.

Prerequisite. Run the **Table Analysis** first (it fills `/SNP/CE00TABINFO`). Data objects cannot be processed without it.

04 Following the Data

The journey of a record in six steps

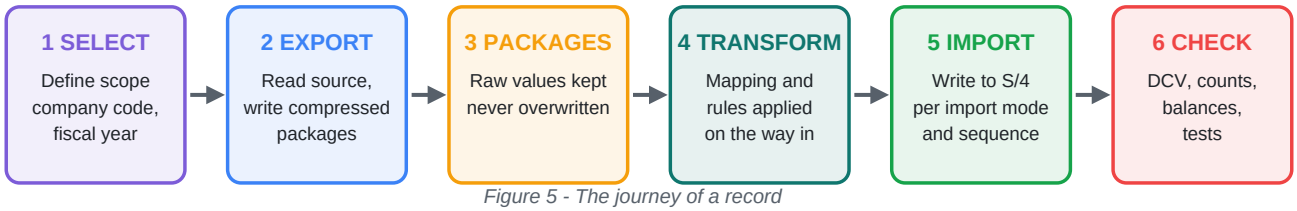

Figure 5 - The journey of a record

Step	What happens	Why it is done this way	What can go wrong
0 Prepare	Table analysis, validation of selections, mappings and objects, DDIC analysis, syntax check.	Errors found before a run cost minutes; found in a run they cost hours.	Skipping validation, then failing mid-run on a missing mapping.
1 Select	Selections limit which rows are read: global selections by domain, or field selections on one table.	Reusable scope: one company code selection works in every table with that field.	A field selection with no value exports the whole table; a global selection with no value transfers no records.
2 Export	Export programs on the source read by selection and write packages.	Source is only read, so the export can be repeated.	Source locks or load on busy systems; wrong reader for a huge dependent table.
3 Packages	Compressed files stored per scenario, block and table. A restart exports only failed tables.	Decouples source from target; allows re-import without re-export.	Deleting a block deletes its files; choosing the wrong export for the import.
4 Transform	Value mappings, global and field rules, events, derivations.	Business changes are applied once, centrally, and are traceable.	Unmapped value; a rule that skips the wrong record.
5 Import	Programs on the target read the packages and write per import mode and sequence.	Mode decides behaviour for keys that already exist.	Insert onto existing keys terminates; Modify, Update or Delete used without agreement.
6 Check	Compare counts and keys, foreign key checks, balance comparison, application tests.	Proves completeness and consistency for sign-off.	Comparing without accounting for mapped values.

4.1 What runs where

Block type	Runs on	Plain meaning
EXP export	Source	Pack the data.
FOR forerunner	Target	Rehearsal that only builds mapping entries; nothing is written to the data tables.
IMP import	Target	Unpack and write the data.
UPL upload	Target side	Load a CSV file into a table.
DEL deletion	Target	Clean-up block created from an import block after successful execution.

4.2 Running options in plain words

Option	Plain meaning	When
Simulation	Do everything except write to the database.	Before the first real import of a block.
Restart	Continue a failed block from the last good package.	After fixing the cause.
Reset	Clear status and intermediate results.	Only when a complete re-run is intended.
Background mode	Run as a background job.	Production; dialog only for testing and debugging.

05 The Three Levers

Selections, mapping and rules: how you steer the result

5.1 Selections: what goes

Type	Plain meaning	Use when
Global selection	One list of values for a field type, for example company codes; applies to every table with that field.	Scope that must be identical everywhere.
Field selection	A condition on one field of one table.	A special filter for a single table.
Declarative (default)	Conditions are combined with AND.	Normal case, recommended.
Free / SQL expert	You write the condition, including OR and NOT.	Complex cases only; switching back to declarative loses assigned selections.

Ranges use four parts: sign, option, low, high. Examples from the SNP guide: `IEQ1000` include equal 1000; `ICP100*` include pattern; `IBT10002000` include between; `EEQ9999` exclude equal.

5.2 Mapping: what changes

A mapping table translates a source value into a target value. Source data is never changed; the translation happens during the transformation.

Pattern	Plain meaning	Example from the guide
1:1	One source value to one target value.	WERKS 1000 to 1001
N:1	Several source values merge into one target value.	WERKS 1000 and 2000 to 2001
1:N	One source value to several targets. Not allowed: it causes an error.	WERKS 1000 to 1001 and 2001
X+N:1 (dependent)	Two source fields together decide one target value.	WERKS + LGORT to LGORT

Mapping layer	Plain meaning
Global mapping	One rule for every field of the same domain, data element or table field.
Field mapping	Only one table field; overrides the global mapping.
Value mapping	Known value to known value; wildcards and ranges; bulk CSV upload.
Derivation	ABAP logic works out the target from other fields.
Hard-coded / default	Always a fixed value / a value only when the source field is initial.

Remember. More specific mapping entries win over wildcards. A field that has no import rule is transferred 1:1. Values are case-sensitive. Validate mappings before every run with `/SNP/TE02_VALIDATE_MAPPINGS`.

5.3 Rules and events: what logic runs

Rules are small ABAP routines that fire at defined moments, called **events**. Export events run on the source side; import events on the target side. A global rule applies to every field of a domain; a field rule overrides it for one field.

Moment	Typical use
Before / after table	Prepare or finish something for the whole table, for example clear an own table.
Before / after package	Work on a batch of records, for example insert collected values.
Before / after data record	Inspect or change one record; set EV_SKIP to skip it.
Global export / import rule	Change a value for every field of a domain, for example the client or the logical system.

Changes made in an earlier event are visible in later events of the same cycle. See the Worked Examples for a rule in action.

06 The Tricky Topics, Simply

Each one: in plain words, why it exists, what to remember

Topic	In plain words	How it works	Remember
Number ranges	Documents have numbers from a number range. If the target already uses numbers, copying old numbers could clash.	Strategy per range: Increment (new numbers via NUMBER_GET_NEXT, mappings record old and new), Difference calculation (new = old + an NRIV offset, much faster), 1:1 for internal ranges (numbers adopted), Inactive for external ranges.	Save NRIV before, Update NRIV before import. API_NUMBER_GET_NEXT returns sy-subrc 0 when the offset calculated the value and 4 when NUMBER_GET_NEXT generated it.
Forerunner (FOR)	A rehearsal block that only builds the mapping for key fields, so later blocks that point to those keys can use the mapping.	It runs after the export and before the import, imports in simulation mode with one process per file and writes nothing to the data tables.	The linked objects must have been exported successfully.
Duplicate Key Prevention (DKP)	A safety switch that stops duplicates when the parent selection table has more key fields than the connection uses.	It forces one process per table and logs every exported record, so it is slow.	Avoid it with the Direct Reader, a corrected object structure, or a unique selection on the missing key field.
Import modes	What to do when the key already exists in the target.	Insert (error), Insert ADK (skip), Update (change non-key fields only), Modify (update or insert), Delete (remove by key).	Agree with the business which records are kept before using anything other than Insert. See Worked Example 4.
Address object	Addresses are shared by many objects, so they are collected by number.	Address numbers are collected by global export rules for domains such as ADRNR; export the address object last and import it first.	Wrong order leaves objects pointing to addresses that do not exist yet.
Near-zero downtime (RBNZD)	Most data is moved while the source is still live.	Initial load, then several delta loads that shrink each time, then a short final delta at cutover. Triggers on the source record the changes.	Triggers must be on before the initial load; lock number ranges before each delta import.
Staging manager	A way to prepare or check data in the target without disturbing live tables.	Type D (DDIC): shadow tables switched by renaming. Type C (client): clients AAA to ZZZ.	Use cases include downtime minimisation, recycle bin, foreign key check without writing.

Training decks versus guide. Several details (forerunners, number-range strategies, offsets) are shown in the 19.03 training decks. Check release-specific behaviour against the 25.09 user guide before relying on it.

07 Decision Guide

Which option should I choose?

Situation	Start with	Switch when	Why
Normal parent and child tables	Scout Reader, import mode Insert	A huge dependent table is slow	Default is restartable per table; Direct Reader uses a join but one process and full re-export.
Selection scope for many tables	Global selection	One table needs a special filter	Global is reusable; field selection is local.
Org unit changes (company code, plant)	Global mapping object	One field behaves differently	Field mapping overrides global.
A value needs logic	Derivation or field rule	Pure lookup is enough	Lookups are easier to maintain than code.
Target already has documents	Number-range strategy Difference calculation	Offset result is not valid	Faster; falls back to NUMBER_GET_NEXT.
Mapping needed before the main import	Forerunner object	No linked key domains	Builds mapping, writes no data.
First import of a block	Simulation	After simulation is clean	No data written; shows key and duplicate problems.
Records may exist in target	Insert ADK or Modify (by agreement)	Business wants an overwrite or skip	Insert would terminate on duplicates.
Test again after a failure	Restart	Cause needs a full re-extraction	Restart continues from the last good package; Reset clears status.
Large volume, short downtime	RBNZD delta loads	Volume is small	Only the last delta needs downtime.
Back up target tables before transformation	RET scenario (save, reset, restore)	Not needed	Source equals target; export is the backup, restore imports it with mode INSERT.

This is a decision aid built from the SNP documents. It is not a mandatory SNP procedure; confirm choices for Sempra objects in the design workshops.

7.1 Questions to ask before any run

#	Question	If the answer is no
1	Is the table analysis current?	Run /SNP/TE02_UPDATE_TABLE_INFO.
2	Are selections, mappings and objects validated?	Run the three validation reports.
3	Is the syntax check clean?	Run /SNP/TE02_API_EXEC_SYNTAX_CHK.
4	Do I know which export the import will read?	Select the export explicitly.
5	Do I know what the import mode does to existing keys?	Check section 6 and Example 4.
6	Are number ranges saved and updated?	Save NRIV, then Update NRIV.
7	Is simulation done for new or changed blocks?	Simulate first.

08 Common Mistakes

What goes wrong and the habit that prevents it

Mistake	What happens	Habit that prevents it
Empty field selection assumed to mean nothing	A field selection with no value exports the whole table.	Check every field selection before each export.
Empty global selection assumed to mean everything	A global selection with no value transfers no records.	Open the selection overview before the run.
Unmapped source value	Import fails with no target value found.	Validate mappings; add the value; regenerate blocks.
Switching free selection back to declarative	All assigned selections are lost.	Decide the mode early; document it.
Using Insert when keys exist	Import terminates with a duplicate key error.	Choose the mode by agreement; simulate.
Using Modify, Update or Delete casually	Unintended overwrite or loss in the target.	Restrict to named blocks; get sign-off.
Ignoring the table analysis	Data objects cannot be processed.	Make it a standing prerequisite.
Closing a ticket as ignored	Status becomes Erroneous; later tasks may fail.	Fix or document; use FIXED or NO_CHANGE with a comment.
Forgetting that deleting a block deletes its files	Export no longer available for import.	Copy runs for a new cycle instead of deleting.
Debugging with parallel processes	The debugger does not stop.	Debug in dialog with one process.
Cleaning up after deletion carelessly	The transformation tree is the transformation history.	Clean up only after the deletion is verified.
Skipping the RFC whitelist	Connection failures at execution.	Maintain /SNP/TCSYS on all systems and clients; test the connection.

8.1 How to think when something fails

- **Where did it stop?** Mission Control task, cockpit run, block, or table. Use the log tab and /SNP/TCPROT.
- **Was it the data or the setup?** Setup problems show in the validation reports; data problems show as table-level errors.
- **Can I restart?** Restart continues from the last good package. Reset clears status; use it only when a full re-run is intended.
- **What changed since the last good run?** Mapping, selection, rule or a new source value. Re-validate before re-running.
- **Who must know?** Tickets in Mission Control block the task until resolved; make sure the owner sees them.

09 Glossary

Plain-language definitions in alphabetical order

Term	Meaning
Alias (system)	A short unique name for a system in a scenario; needed when more than one source system is used.
API call	A function you can call inside a rule, for example API_MAPPING_ADD or API_SELECTION_ADD.
Block	The executable unit of a run. Types EXP, FOR, IMP, UPL, DEL.
Bluefield	Selective migration of chosen entities into a S/4HANA system; combines Greenfield and Brownfield ideas.
Brownfield	Converting the existing system to S/4HANA.
Control system	The system that hosts Mission Control, the cockpit, the rule base and the shadow DDIC.
Data object	A business entity at table level, including its table hierarchy.
DCV	Data Consistency Verification: foreign key and fixed value checks after the load.
DDIC	SAP Data Dictionary; the shadow DDIC is a copy used for generating programs.
Derivation	A rule that works out a target value from other fields.
Direct Reader (DR)	Export method using a join; independent of the parent export.
DKP	Duplicate Key Prevention; a safety mechanism that slows the export.
Event	A defined moment (before table, after record ...) where a rule can run.
Export / import	Writing source data to packages / reading packages into the target.
Forerunner (FOR)	A block that builds mapping entries without writing data.
Global selection	A reusable list of values applied to all tables with a matching field.
Greenfield	A new S/4HANA implementation; usually master data and open items only.
Mapping	Translation of source values into target values.
Mission Control	Project and task orchestration tool (/N/SNP/MC).
NRIV	SAP number range table; CrystalBridge saves, checks and updates its levels.
Package	A compressed export file with a limited number of records or size.
RBNZD	Rule-Based Near-Zero Downtime.
RET scenario	A scenario where source equals target, used to save, reset and restore tables.
RFC whitelist	List of permitted RFC connections, maintained in /SNP/TCSYS.
Run	A named execution plan made of blocks.
Scenario	The complete migration blueprint.
Scout Reader (SR)	Default export method using FOR ALL ENTRIES.
Simulation	Execution without writing to the database.
Staging	Holding data in packages (or shadow tables) between export and import.
Ticket	A Mission Control record of a warning or error that blocks the task until resolved.
Transformation Cockpit	The engine room tool (/N/SNP/TC).
Unit	A Mission Control project template.
XPRA / XCLA / SDM	Post-processing programs on the target after import.

10 Self-Check

Test yourself, then see where to go next

#	Question	Answer
1	Is the ECC source changed by a migration?	No. It is only read; all changes happen between export and import.
2	Where does data wait between export and import?	In compressed export packages on the file system; files are never overwritten.
3	What links a Mission Control task to a cockpit scenario?	Configuration: the scenario ID in variable TC_SCENARIO and the API report parameters P_SCEN, S_RUN, S_BLOCK. No linking table is named in the SNP documents.
4	Name the five block types.	EXP, FOR, IMP, UPL, DEL.
5	What does a field selection with no value do?	It exports the whole table. A global selection with no value transfers no records.
6	Why validate mappings before a run?	To find source values without a target value; otherwise the import fails.
7	What is a 1:N mapping?	One source value with several targets. It causes an error.
8	Which import mode skips existing keys without error?	Insert ADK.
9	What does a forerunner write to the data tables?	Nothing. It only creates the mapping.
10	What does API_NUMBER_GET_NEXT return in sy-subrc?	0 when the value was calculated from the NRIV offset, 4 when NUMBER_GET_NEXT generated it.
11	In which order are addresses handled?	Export the address object last, import it first.
12	What shrinks the downtime in RBNZD?	Repeated delta loads; only the final delta runs during cutover.

10.1 Where to go next

If you want to ...	Go to
Practise with concrete values	Worked Examples PDF (mapping, selections, import modes, number ranges, rules, offsets, RET, Mission Control, troubleshooting).
Look up a report, role, parameter or default	Reference PDF, sections 2 to 14.
Fill in the Sempra landscape	Reference PDF, section 2.3 and 2.4 (all fields are **TBC**).

10.2 Open items

#	Open item	Status
1	Sempra landscape: SIDs, releases, clients, volumes, company codes, dates	**TBC**
2	Whether carve-out content (Module 5 title) should be emphasised for Sempra	**TBC**
3	Further SNP documents to fold in	**TBC**

Sources: Kyano Mission Control Complete Reference Guide; CrystalBridge Transformation User Guide 25.09 SP00; Module 5 Transformation Advanced (v25.03); Transformation Cockpit SNP2 and Advanced SNP3 training decks (19.03 SP00). Analogies are teaching aids. Document generated with IBM Bob.